

درس ابزار دقیق

حسگرهای هوشمند و حسگرهای نرم و ابزار مجازی

Smart Sensor, Soft Sensor,
Virtual Instrument

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✓ حسگر هوشمند

✓ حسگر نرم

✓ ابزار مجازی

○ معرفی نرم افزار LABVIEW

The term **smart sensor** was coined in the mid-1980s and, since that time, several products have been referred to as smart sensors. The intelligence required by these devices is available from microcontroller (MCU), digital signal processor (DSP), application specific integrated circuit (ASIC), or field programmable gate array (FPGA) technologies developed by several semiconductor manufacturers.

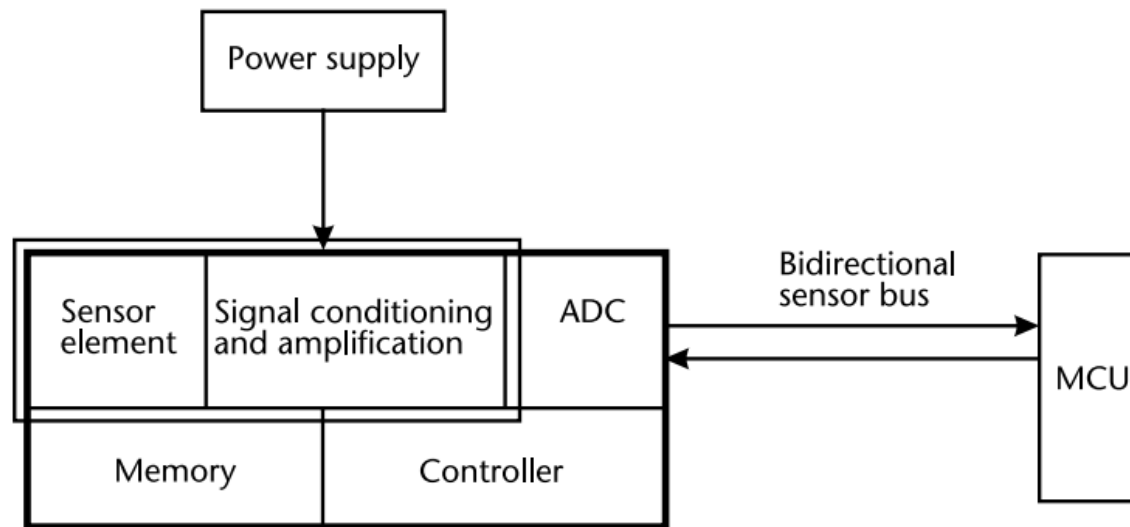


Figure 1.5 Smart sensor model.

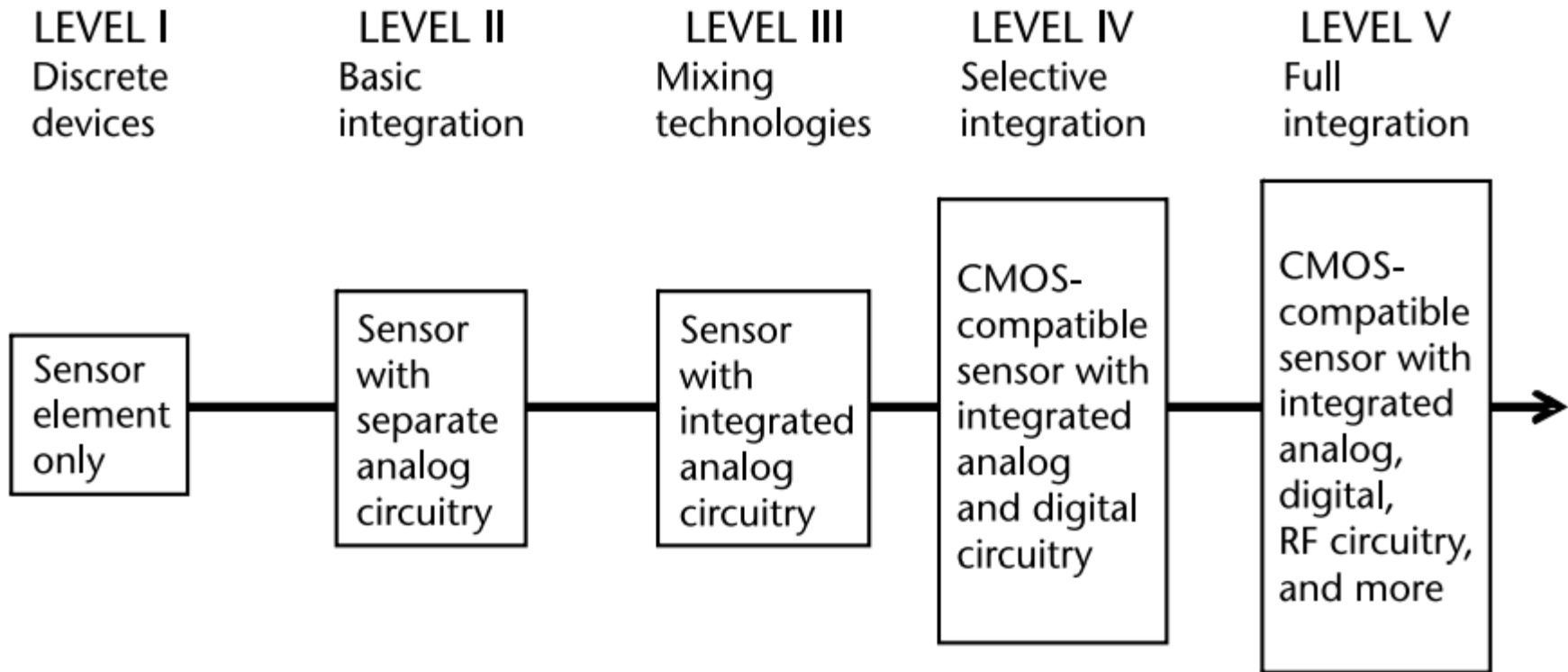


Figure 1.7 Sensor technology migration path.

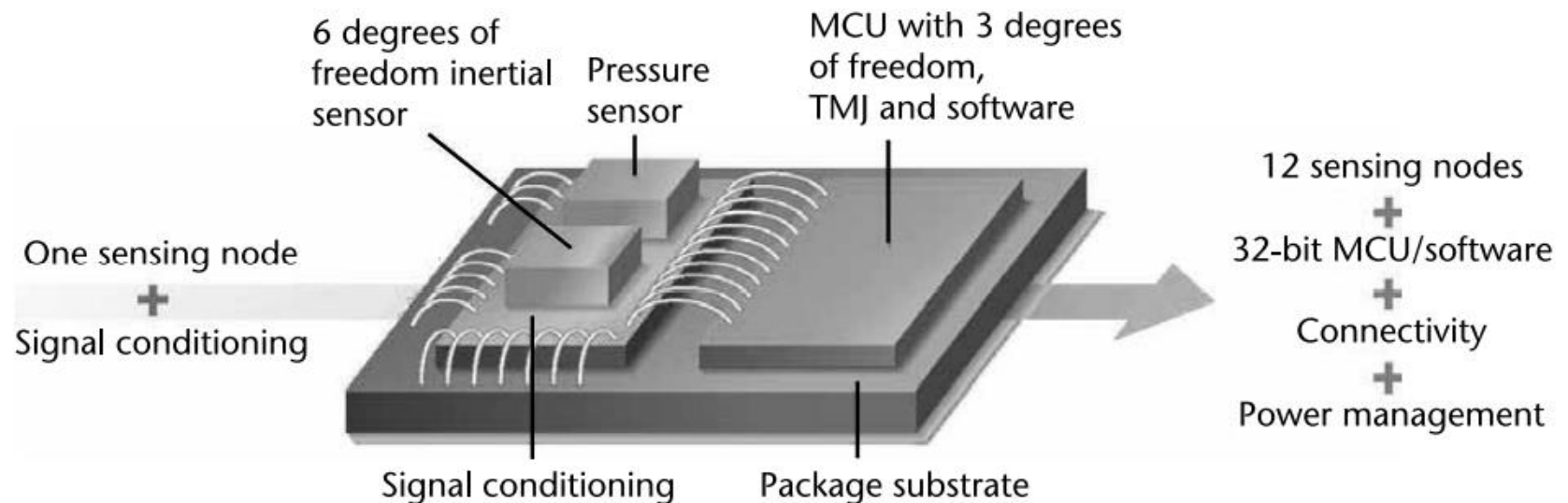


Figure 1.10 A multichip smart sensor for acceleration and pressure measurements. (Courtesy of Freescale Semiconductor.)

MPL3115A2

I²C precision pressure sensor with altimetry

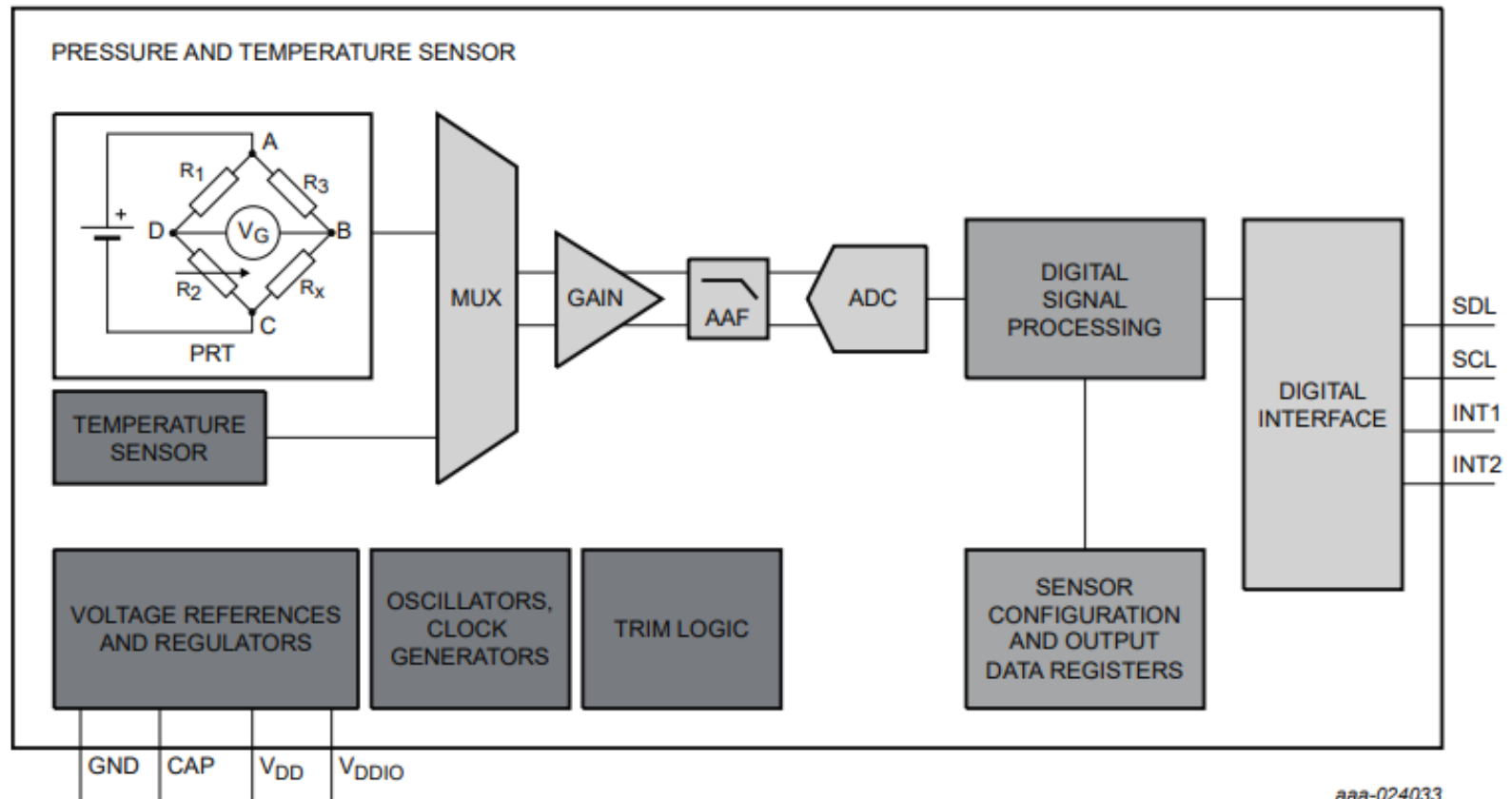


Figure 1. Block diagram

Soft sensors are a valuable tool in many different industrial fields of application, including refineries, chemical plants, cement kilns, power plants, pulp and paper industry, food processing, nuclear plants, urban and industrial pollution monitoring, just to give a few examples. **They are used to solve a number of different problems such as measuring system back-up, what-if analysis, real-time prediction for plant control, sensor validation and fault diagnosis strategies.**

حسگر نرم – لزوم استفاده

Potential applications of soft sensors can be imagined in any industrial field. Of course some characteristics (**large plants, hard-to-measure process parameters, measurement delay, working environments hostile to measuring device survival, hard control requirements,** etc.) of the plants considered can encourage the application of this technique.

ویژگی های جذاب حسگر نرم:

Soft Sensors offer a number of attractive properties:

- 1- They represent a low-cost alternative to expensive hardware devices, allowing the realization of more comprehensive monitoring networks;
- 2- They can work in parallel with hardware sensors, giving useful information for fault detection tasks, thus allowing the realization of more reliable processes;
- 3- They can easily be implemented on existing hardware (e.g. microcontrollers) and retuned when system parameters change;
- 4- They allow real-time estimation of data, overcoming the time delays introduced by slow hardware sensors (e.g. gas chromatographs), thus improving the performance of the control strategies.

رویکردهای متفاوت در طراحی حسگر نرم:

There are three main approaches to building soft sensors:

- 1-Mechanistic modeling (physical modeling),
- 2- Multivariate statistics,
- 3- Artificial intelligence modeling such as neural networks, fuzzy logic and hybrid methods.

Collected data can be processed in different ways to design the soft sensor. A number of choices are necessary in order to select both the **model class** (e.g. linear or nonlinear, static or dynamic, and so on) and the **identification** approach most suitable to the problem under investigation. The last step in soft sensor design, i.e. the problem of **model validation**, can be approached using a number of different strategies.

soft sensor design and can be summarized as follows:

- 1- Data collection and filtering;
 - 2- Variables and model structure selection;
 - 3- Model identification;
 - 4- Model validation.
- Model class (e.g. linear or nonlinear, static or dynamic)

VIs can be considered as a wider class than soft sensors.

In fact, **soft sensors focus on the process of estimation** of any system variable or product quality by **using mathematical models, substituting** some physical sensors and using data acquired **from some other** available ones.

For their part, **VIs are based on software** that performs any of the typical actions involved in a measurement and/or control problem, by exploiting available instrumentation, computers and software. This action can either involve, or not, modeling capabilities typical of soft sensors.

VIs are the result of the rapid diffusion that has taken place in the last 20 years of low-cost Windows-based personal computers, Macs, and workstations in any engineering application field, along with performing software.

ابزار مجازی همچون ابزار دیگر از سه قسمت تشکیل شده است: اندازه گیری، محاسبات، رابط کاربری

Three main functional blocks can be recognized in any instrument, and all of them have been affected by changes introduced by the introduction of VIs:

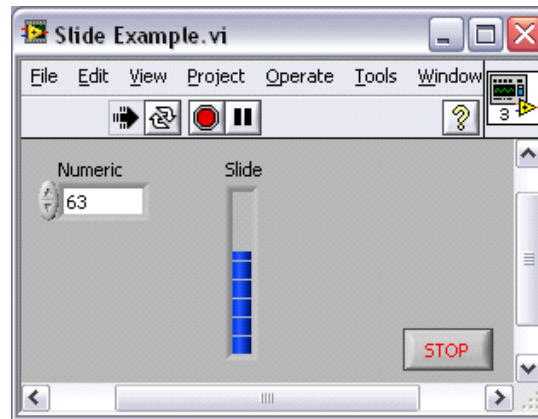
- 1- Measurement;
- 2- Computation;
- 3- User interface.

LabVIEW Programs Are Called Virtual Instruments (VIs)

Each VI has 2 Windows

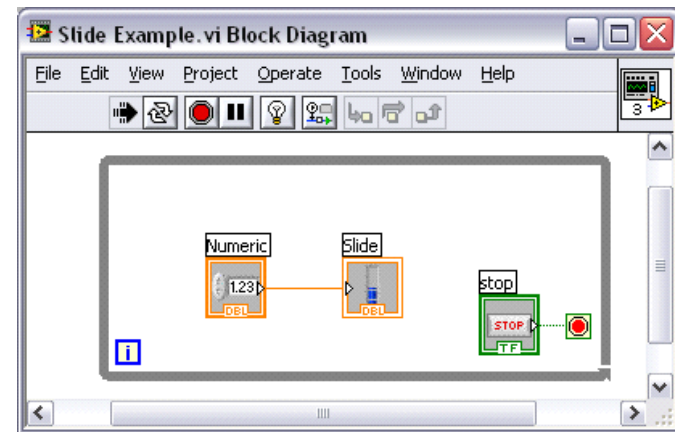
Front Panel

- User Interface (UI)
 - Controls = Inputs
 - Indicators = Outputs

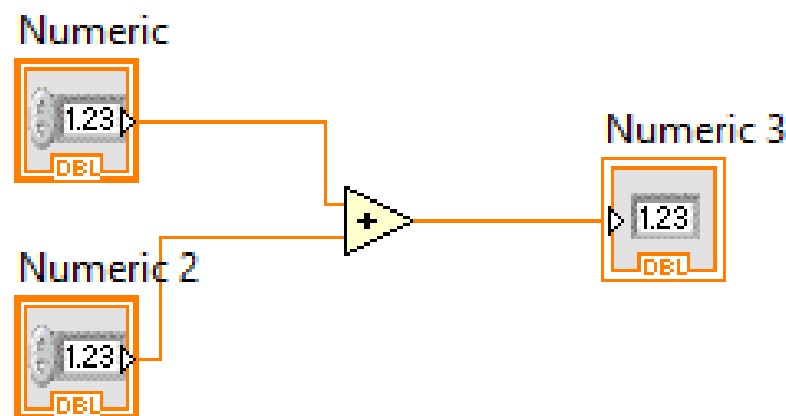
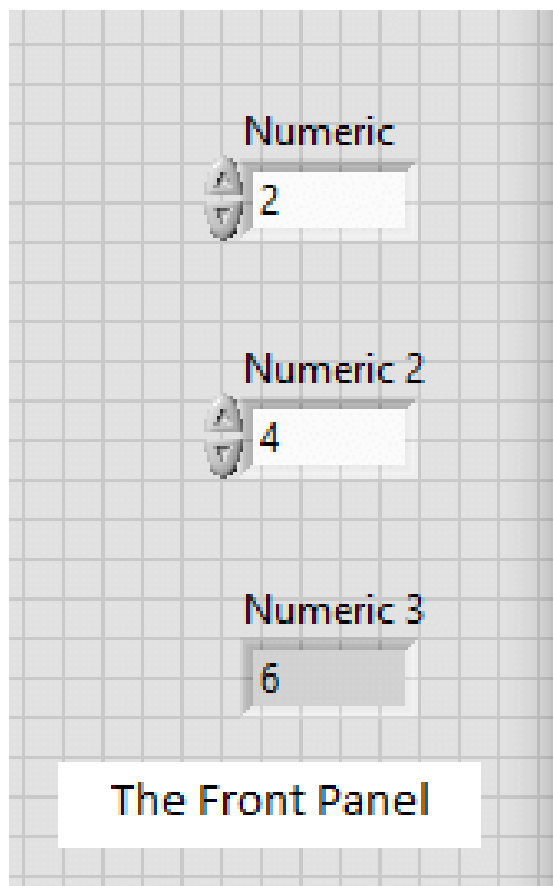


Block Diagram

- Graphical Code
 - Data travels on wires from controls through functions to indicators
 - Blocks execute by Dataflow



مثال از پنل جلویی و پنل پشتی (بلوک دیاگرام) در نرم افزار LABVIEW:



The Block Diagram



36pt Application Font



Signal Generation

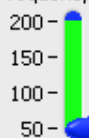
Input Signal 1

Input Signal 2

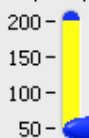
square

sine

Frequency (Hz)



Frequency (Hz)



Signal Processing

Select Window

Select Filter

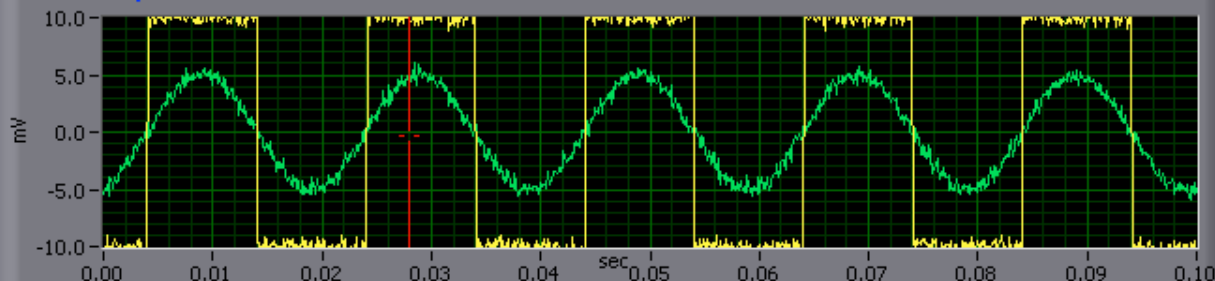
Hanning

Chebyshev

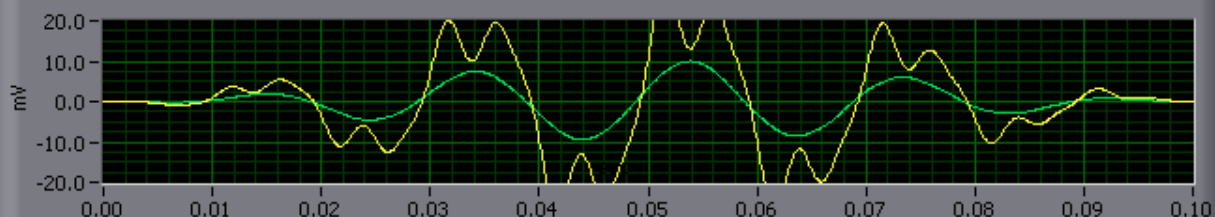
STOP [F4]

MORE INFO...
[F5]

Acquired Waveform

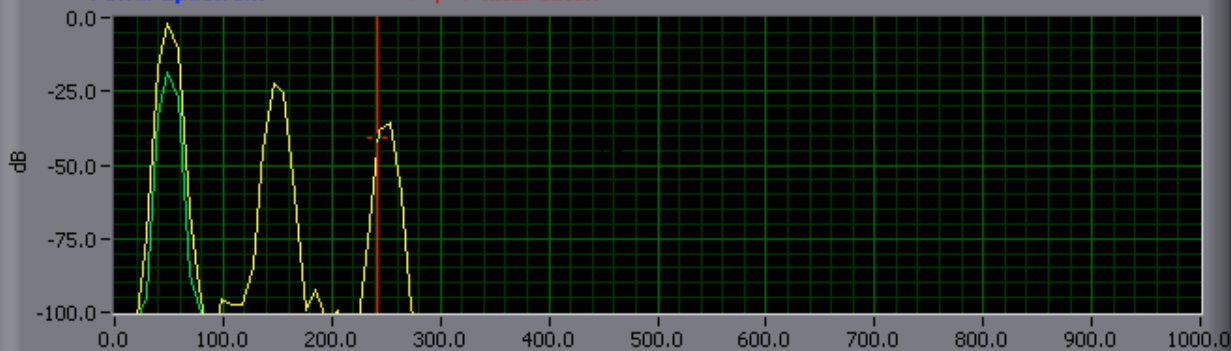


Processed Waveform



Power Spectrum

<-- | --> filter cutoff





Tools Palette



- Floating Palette
- Used to operate and modify front panel and block diagram objects.



Automatic Selection Tool



Operating Tool



Positioning/Resizing Tool



Labeling Tool



Wiring Tool



Shortcut Menu Tool



Scrolling Tool



Breakpoint Tool



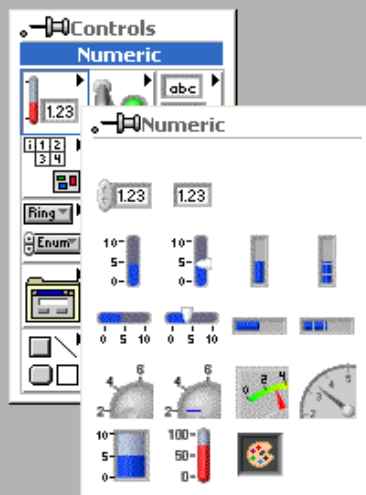
Probe Tool

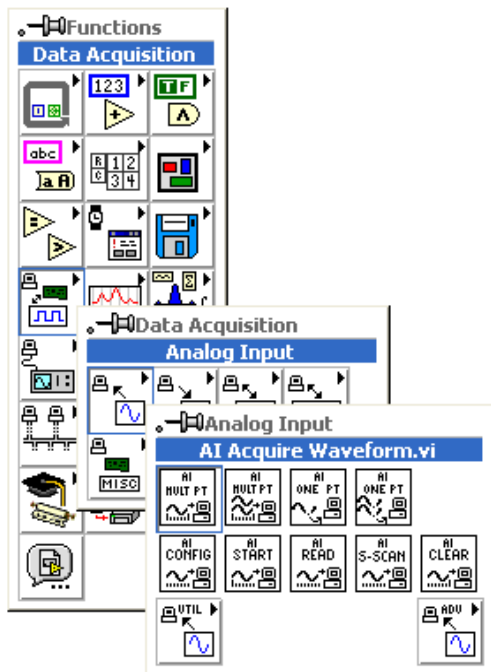


Color Copy Tool



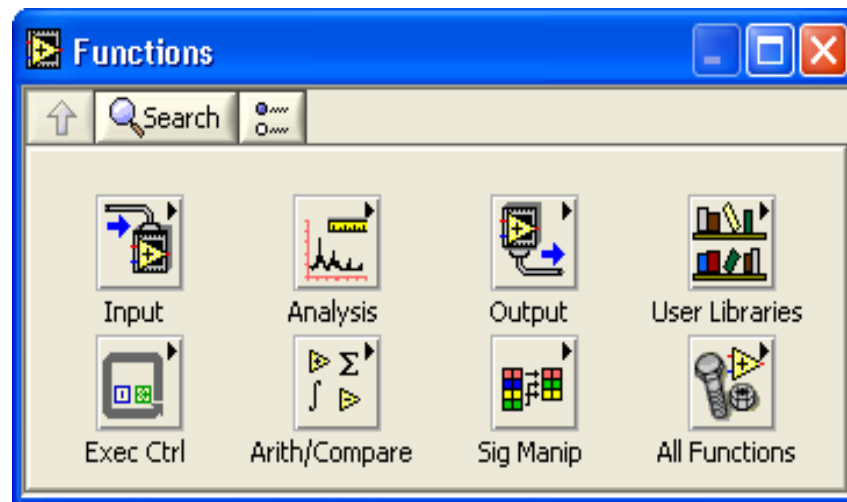
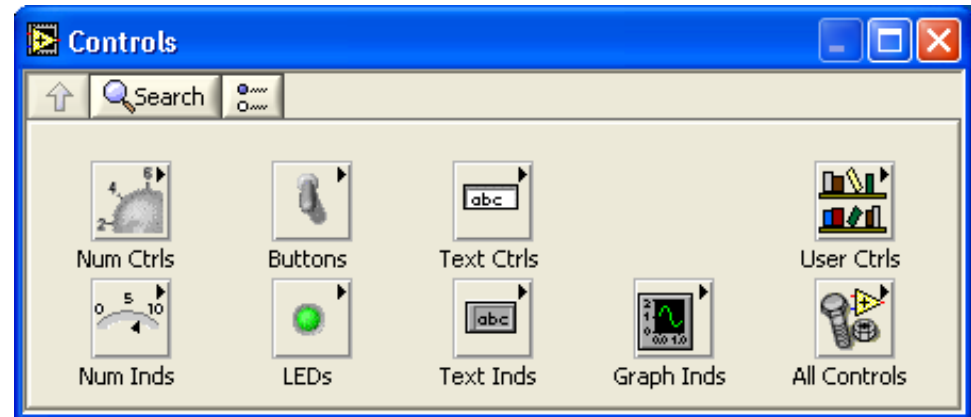
Coloring Tool





Controls and Functions Palettes

Controls Palette (Front Panel Window)



Functions Palette (Block Diagram Window)